Topic : Swift Fundamentals: Variables and Constants

Task 1: Student profile card

```
let studentName = "Asha"
let rollNumber: Int = 2301
var semester = 5
var cgpa: Double = 8.2
let isHosteller = true
print("Name: \(studentName), Roll: \(rollNumber), Semester: \(semester)")
semester += 1
cgpa = 8.5
print("Name: \(studentName), Roll: \(rollNumber), Semester: \(semester), CGPA: \(cgpa)")
print(type(of: studentName))
print(type(of: rollNumber))
print(type(of: semester))
print(type(of: cgpa))
print(type(of: isHosteller))
```

```
Name: Varun, Roll: 2301, Semester: 5
Name: Varun, Roll: 2301, Semester: 6, CGPA: 8.5
String
Int
Int
Double
Bool
```

Task 2: Temperature converter

```
import Foundation

let multiplier = 9.0 / 5.0

let offset = 32.0

let kelvinOffset = 273.15

var celsius = 25.0

var fahrenheit = celsius * multiplier + offset

var kelvin = celsius + kelvinOffset

print("Celsius: \(String(format: "%.1f", celsius))")

print("Fahrenheit: \(String(format: "%.1f", fahrenheit))")

print("Kelvin: \(String(format: "%.1f", kelvin))")

celsius = 100.0

fahrenheit = celsius * multiplier + offset

kelvin = celsius + kelvinOffset

print("Celsius: \(String(format: "%.1f", celsius))")

print("Fahrenheit: \(String(format: "%.1f", fahrenheit))")

print("Kelvin: \(String(format: "%.1f", kelvin))")
```

```
Celsius: 25.0
Fahrenheit: 77.0
Kelvin: 298.1
Celsius: 100.0
Fahrenheit: 212.0
Kelvin: 373.1
```

Task 3: Fix the compile errors

```
var price = 250

var quantity: Int = 3

let taxRate = 0.18

price = 300
```

```
let subtotal = Double(price * quantity)

let tax = subtotal * taxRate

let total = subtotal + tax

var title = "Order"

title = "Order 42"

print(title)

print("Subtotal: \(subtotal)")

print("Tax: \(tax)")

print("Total including tax: \(total)")
```

```
Order 42
Subtotal: 900.0
Tax: 162.0
Total including tax: 1062.0
```